# 1.7.8: Crystal Structure

The structure of a crystal can be described by combining the following elements: the lattice type, the lattice parameters, and the motif.

The **lattice type** defines the location of the lattice points within the unit cell.

The **lattice parameters** define the size and shape of the unit cell.

The **motif** is a list of the atoms associated with each lattice point, along with their fractional coordinates relative to the lattice point. Since each lattice point is, by definition, identical, if we add the motif to each lattice point, we will generate the entire structure:

Plan view

Knowing the motif and lattice it is possible to construct a **Plan view** of the crystal structure. The **Plan view** is the standard representation of a crystal structure and is very easy to produce. It is generally the 2D projection looking down the [001]/z-axis of the unit cell. Note this is equivalent to constructing a projection on the (001) plane. Refer to Lattice Planes and Miller Indices TLP for information on lattice planes.

The Plan view generally displays a 2×2 array of unit cells. The heights of the atoms within the unit cell are represented by fractions next to them, the fraction indicating that atom's fractional height in terms of the unit cell height (c) (atoms at the top and bottom of the unit cell have no numbers next to them). On constructing the plan view it is essential to not only indicate heights of atoms within the unit cell but also define the crystallographic axes you are using along with tracing out the unit cell.